CMPn332 – CMPN332 – CMP432

What's an Image



- Image is a 2D projection of 3D objects.
- So, image can be defined as 2D function F(x,y)
  Where:
  - x,y represents spatial coordinates
  - The amplitude of F at any point is called the intensity at that point.

Image Types

- Analog Image and Digital Image





Analog Image→ continuous levels of coordinates and intensities
Digital Image→ **finite set of digital values, called picture elements or pixels**

Why do we need image processing

- Improve the visual effect of people by improving the quality of the image.
- Machine/Robot vision
- Medical field
- Efficient storage & transmission.

So digital image processing is:

- Set of algorithms applied to digital images in order to improve quality, storage, transmission and representation.

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Needed environment

- Hardware:
  - CPU (Intel i5/ i7/ Xeon recommended).
  - 8 GB RAM, 10 GB HDD Free Space.
- Software
  - Opencv → latest release
  - Visual studio → latest release

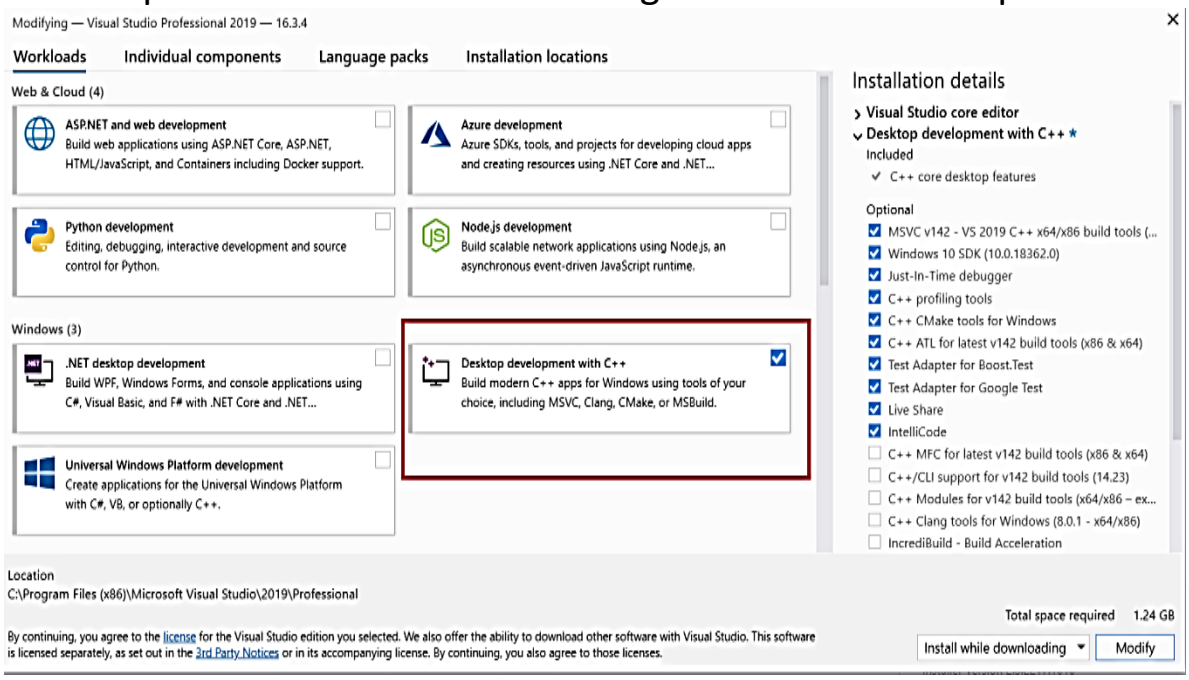
OpenCv

- OpenCv stands for open source library for image processing and computer vision.
- OpenCv supports Windows, Linux, Android and Mac OS.
- OpenCv supports a wide variety of programming languages such as C++, Python, Java, etc..
- Primary interface of OpenCV is in C++
- Opencv Download link→ Here

visual studio

visual studio community free download link→ Here

- There are different kinds of component. What we will need is desktop development with c++ check it and goes with the install process

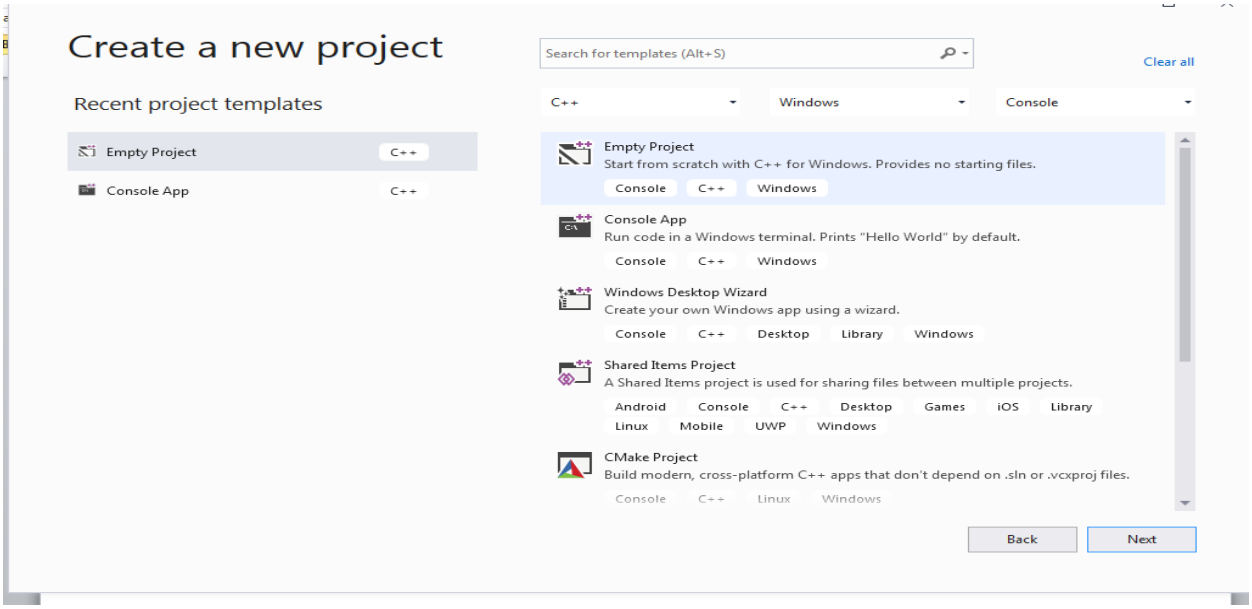


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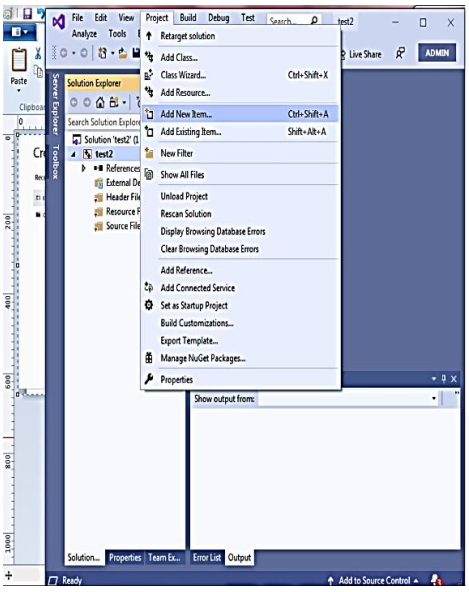
Add opencv dll files path

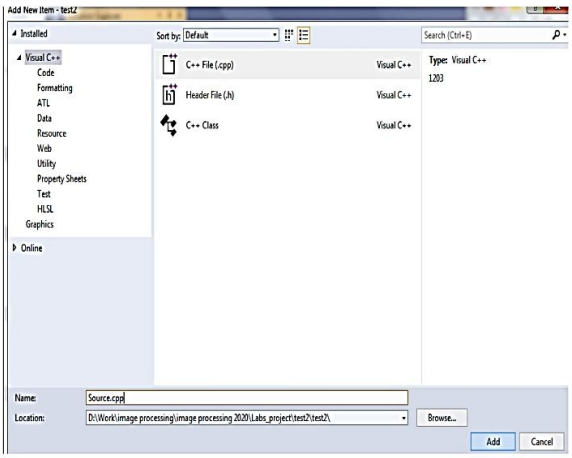
1. environment variables →system variables →double click on path→ write the path of the bin folder on your device. for example “E:\opencv\build\x64\vc15\bin”

2. visual studio →create new project



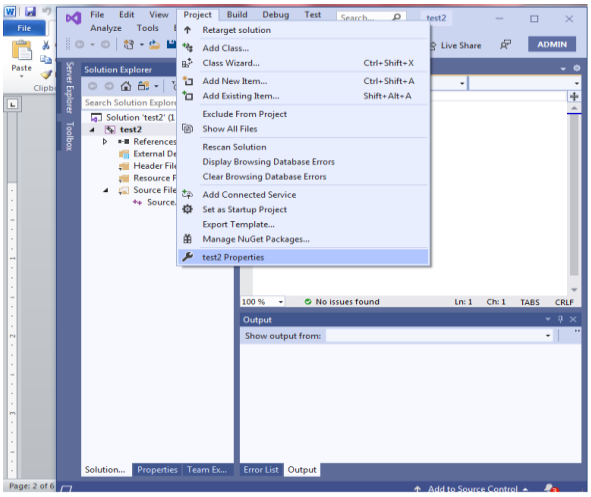
3. Add C++ file

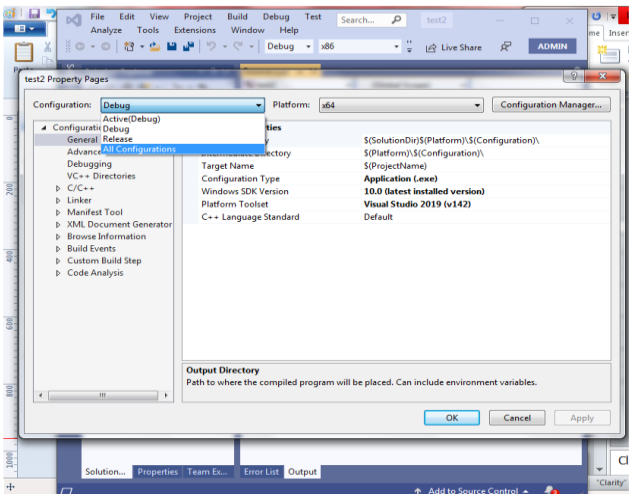




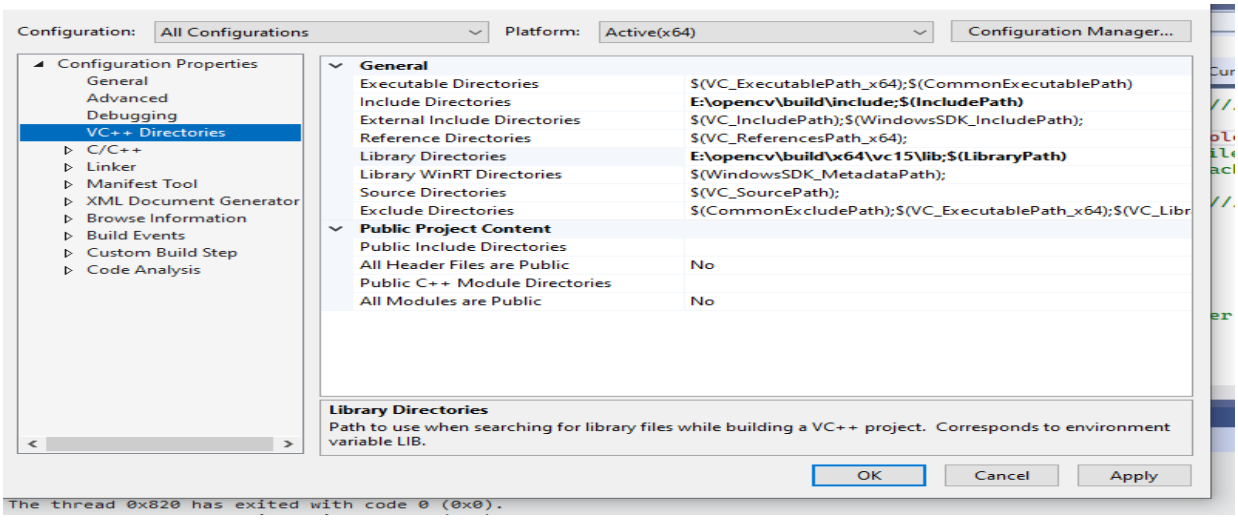
4. Add opencv files to your project

- Project→ properties
- **Configuration→ all configuration and platform→x64**



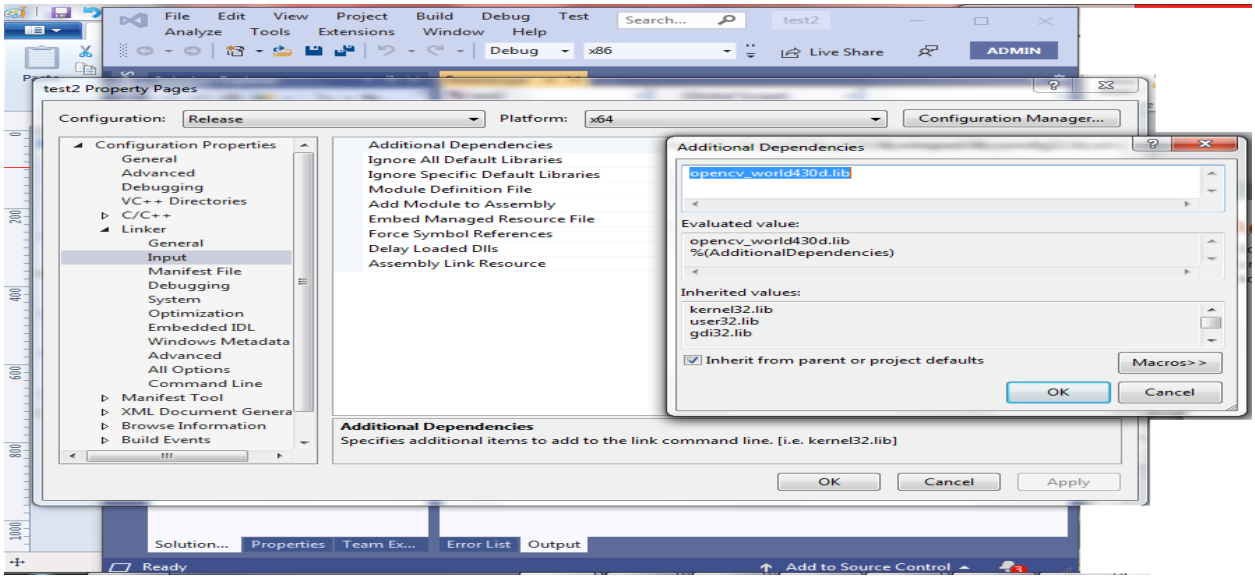


  - VC++ Directories →General
    - Include Directories→write the path of the include folder on your device. For example“**E:\opencv\build\include”**
    - Library Directories→ write the path of the lib folder on your device. for example “**E:\opencv\build\x64\vc15\lib”**



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- **Configuration→ Release and platform→x64**
  - Linker→ input
    - Additional dependencies→opencv_world430.lib



Display image

```
#include<opencv2/opencv.hpp>
#include"iostream"
using namespace std;
using namespace cv;

int main()
{
	Mat image, gray_image;
	image = imread("C:\\Users\\Public\\Pictures\\Sample Pictures\\xx.jpg");
		//image = imread("xx.jpg");
	//Check for invalid input
	if (image.empty())
	{
		cout << "no image" << endl;
		system("pause");
		return -1;
	}
	namedWindow("RGB image", 0);
	imshow("RGB image", image);
	cvtColor(image, gray_image, COLOR_BGR2GRAY);
	namedWindow("gray image", 0);
	imshow("gray image", gray_image);
	imwrite("gray_image.jpg", gray_image);
	waitKey(0);

}
```

